

5(a)	work done per unit charge	B1
	work done (on charge) in moving positive charge from infinity (to the point)	B1
5(b)(i)	radius = 0.060 m	A1
5(b)(ii)	$V = Q / 4\pi\epsilon_0 x$ $Q = (-) 850 \times 4\pi \times 8.85 \times 10^{-12} \times 0.060$ or $Q = (-) 850 \times 0.060 / 8.99 \times 10^9$ <i>(any correct pair of V and x values from curve)</i>	C1
	$Q = -5.7 \times 10^{-9} \text{ C}$	A1
5(c)(i)	$E_P = Q^2 / 4\pi\epsilon_0 x$ $= (5.67 \times 10^{-9})^2 / (4\pi \times 8.85 \times 10^{-12} \times 0.46)$	C1
	$= 6.3 \times 10^{-7} \text{ J}$	A1
5(c)(ii)	- force is repulsive so spheres move apart - force in direction of motion so speed increases - potential energy converted to kinetic energy so speed increases - force decreases with distance so acceleration decreases - momentum is conserved (at zero) (and masses are equal) so velocities are always equal and opposite <i>Any three points, 1 mark each</i>	B3